

Extended exercises for Lecture "Theoretical Spectroscopy for systems implying periodic boundary conditions"**Exercise: Vibrational analysis for molecule on slab**

based on exercises taken from the [CP2K Homepage](#).

1. Vibrational analysis for molecule on slab

- After having optimized a slab geometry, read in the XYZ coordinates of the slab with e.g. Avogadro. Add the XYZ coordinates of the molecule to be adsorb on top of the slab.
- Perform a geometry optimization, relaxing only the upper layer of the slab and the molecule, while keeping the lower layers of the slab fixed, thus adding a `&CONSTRAINT` section in the `&MOTION` section


```
&MOTION
&CONSTRAINT
&FIXED_ATOMS
LIST ...
&END FIXED_ATOMS
&END CONSTRAINT
&END MOTION
```
- Subsequently, the computation of vibrational frequencies can again be requested by changing the run type from geometry optimization (`RUN_TYPE GEO_OPT`) to vibrational analysis (`RUN_TYPE VIBRATIONAL_ANALYSIS`) in the CP2K input file as well as by adding a section `&VIBRATIONAL_ANALYSIS` and a print comment (`&PRINT/ &MOMENTS ON`) to switch on the computation of corresponding dipole moments.
- The `&VIBRATIONAL_ANALYSIS` section can thereby be chosen as for the analogous analysis for the slab geometry,


```
&VIBRATIONAL_ANALYSIS
INTENSITIES T
FULLY_PERIODIC TRUE
DX 0.001
&END VIBRATIONAL_ANALYSIS
```
- You can compare and visualize the obtained vibrational frequencies and corresponding intensities using the Fortran code [newspecmatch](#).
- To use `newspecmatch`, insert the following two lines in your `.bashrc` file and source the file:


```
export newspecmatchhome=/zfs/home/suphc356/newspecmatch/src
export PATH=$PATH:$newspecmatchhome
```

 Open the `.bashrc` file with `vim` and append the two lines.
- `Newspecmatch` enables with the option `-plot` to plot the spectral function:


```
newspecmatch spectra_1.dat -plot -fscal 0.982 -range 0 3500.
```

Furthermore, vibrational spectra can also be compared with the norms closer to 1 indicating a good match:

```
newspecmatch spectra_1.dat spectra_2.dat
```

- You can scale the obtained frequencies by adjusting the global scaling factor `fscal` and plot the spectra again using the `plot` keyword and `gnuplot spectra.gnup`.